

Research Statement

Self-Evolving Agents, Long-Term Memory, and Control

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Research focus

I want to study self-evolving and open-ended LLM-based agents: systems that do not only solve isolated tasks, but carry experience forward. Such agents may remember previous interactions, reuse earlier decisions, update strategies, create tools, and change their behavior over time. My focus is long-term memory, because memory is one of the main mechanisms through which this kind of adaptation happens.

I do not see agent memory as only a retrieval module or a larger context window. Memory changes what the agent does next. It can make an agent more useful, but it can also make failures persistent. A wrong memory can be reused many times. A poisoned document can become a trusted reference. A stale assumption can silently influence later decisions. The user may not know what the agent retained, where it came from, or how to correct it.

Central question: How can long-term memory help self-evolving agents adapt over time while keeping their behavior reliable, inspectable, private, and controllable?

Past work and motivation

My M.Sc. thesis at the Max Planck Institute of Colloids and Interfaces focused on missing-data imputation in time series of malaria mosquito counts. I built a two-stage pipeline using MICE and XGBoost with Poisson loss. The hard part was not only building a model. It was deciding what information was valid, avoiding leakage, aligning environmental signals, evaluating on masked known values, and keeping the experiments reproducible.

That work shaped how I think about reliability. A system can produce confident outputs while hiding fragile assumptions. If the evaluation is wrong, the result is misleading. If the data is incomplete or shifted, the model may still look useful. In agents, this becomes more serious because an agent can reuse previous outputs and carry assumptions into later actions.

I also bring industry experience from Siemens Energy and Mercedes-Benz, where I worked with data pipelines, dashboards, data quality, and decision-support systems. These settings taught me that practical systems fail not only when a model is wrong, but also when context is unclear, information is stale, or users cannot trace how an output was produced.

Future directions

- **Memory formation and revision:** How should an agent decide what becomes long-term memory, what remains temporary context, and when a memory should be weakened, updated, or forgotten?
- **Evaluation across time:** How can we test agents over multiple episodes with changing documents, code repositories, scientific data, or knowledge bases, rather than only single-turn tasks?
- **Control and auditability:** How can users and researchers inspect what an agent learned, trace where a memory came from, and understand how it influenced later behavior?

My independent project idea, BrainLLM / Subconscious, is connected to this direction. The idea is a local-first memory layer over files, folders, documents, and user context using LLMs, embeddings, RAG, vector search, and possibly a knowledge graph. For a PhD, I want to treat this as a research problem: how memory-based agents can adapt over time without becoming opaque, unsafe, or difficult to control.

Fit

I am applying to this group because my research interest sits directly at the intersection of self-evolving agents, long-term memory, and real-world safety. I want to understand how agents change when they keep memory across tasks, and how we can evaluate and control that change before such systems become deeply embedded in daily work.